

TU	SOL		LUA				Lat.	CREP		SOL Nascer	LUA - Nascer			
	AHG	Dec	AHG	v	Dec	d Ph		Náut.	Civil		21	22	23	24
d h	o	o	o	o	o	o	o	h m	h m	h m	h m	h m	h m	h m
21	00	176 34.4 S10 49.8	205 22.7 10.5	S21 46.2	4.7	55.3	N 72	05 46	07 04	08 17	09 49	09 17	08 59	08 45
	01	191 34.4 48.9	219 52.2 10.6	21 41.5	4.8	55.3	N 70	05 47	06 58	08 03	09 47	09 17	08 59	08 45
	02	206 34.5 48.0	234 21.8 10.6	21 36.7	4.9	55.2	68	05 48	06 52	07 51	09 27	09 06	08 54	08 45
	03	221 34.6 47.1	248 51.4 10.7	21 31.8	5.0	55.2	66	05 49	06 48	07 41	08 37	08 37	08 36	08 34
	04	236 34.6 46.2	263 21.1 10.7	21 26.8	5.1	55.2	64	05 49	06 44	07 33	08 05	08 16	08 21	08 24
	05	251 34.7 45.3	277 50.8 10.8	21 21.7	5.2	55.2	62	05 49	06 41	07 26	07 42	07 58	08 09	08 16
	06	266 34.8 S10 44.4	292 20.6 10.8	S21 16.5	5.3	55.2	60	05 49	06 38	07 20	07 23	07 44	07 58	08 09
	07	281 34.8 43.5	306 50.4 10.9	21 11.2	5.4	55.1	N 58	05 49	06 35	07 15	07 07	07 31	07 49	08 02
	08	296 34.9 42.6	321 20.3 11.0	21 05.8	5.5	55.1	56	05 49	06 32	07 10	06 54	07 21	07 41	07 57
	09	311 35.0 41.7	335 50.3 11.0	21 00.3	5.6	55.1	54	05 49	06 30	07 06	06 42	07 11	07 33	07 52
EXTA	10	326 35.0 40.8	350 20.3 11.1	20 54.7	5.7	55.1	52	05 49	06 28	07 02	06 32	07 02	07 27	07 47
	11	341 35.1 39.9	4 50.4 11.1	20 49.0	5.8	55.1	50	05 48	06 26	06 59	06 23	06 55	07 21	07 43
	12	356 35.2 S10 39.0	19 20.5 11.2	S20 43.2	5.9	55.1	45	05 47	06 21	06 51	06 03	06 38	07 08	07 34
F E I R A	13	11 35.3 38.1	33 50.7 11.3	20 37.3	6.0	55.0	N 40	05 45	06 17	06 44	05 47	06 25	06 58	07 26
	14	26 35.3 37.2	48 21.0 11.3	20 31.3	6.1	55.0	35	05 44	06 13	06 39	05 34	06 14	06 49	07 20
	15	41 35.4 36.3	62 51.3 11.3	20 25.2	6.1	55.0	30	05 42	06 10	06 34	05 22	06 03	06 41	07 14
	16	56 35.5 35.4	77 21.6 11.5	20 19.1	6.3	55.0	20	05 37	06 03	06 25	05 02	05 46	06 27	07 04
	17	71 35.5 34.4	91 52.1 11.5	20 12.8	6.4	55.0	N 10	05 32	05 56	06 18	04 44	05 31	06 15	06 55
	18	86 35.6 S10 33.5	106 22.6 11.5	S20 06.4	6.5	55.0	0	05 25	05 49	06 10	04 28	05 17	06 03	06 47
	19	101 35.7 32.6	120 53.1 11.6	19 59.9	6.5	54.9	S 10	05 16	05 41	06 03	04 12	05 03	05 52	06 39
	20	116 35.8 31.7	135 23.7 11.7	19 53.4	6.7	54.9	20	05 06	05 32	05 55	03 54	04 47	05 39	06 30
	21	131 35.8 30.8	149 54.4 11.7	19 46.7	6.7	54.9	30	04 52	05 21	05 45	03 34	04 30	05 25	06 20
	22	146 35.9 29.9	164 25.1 11.8	19 40.0	6.8	54.9	35	04 43	05 14	05 40	03 22	04 20	05 17	06 14
23	161 36.0 29.0	178 55.9 11.9	19 33.2	7.0	54.9	40	04 32	05 06	05 34	03 08	04 08	05 08	06 07	
22	00	176 36.0 S10 28.1	193 26.8 11.9	S19 26.2	7.0	54.9	45	04 18	04 56	05 26	02 52	03 54	04 57	05 59
	01	191 36.1 27.2	207 57.7 12.0	19 19.2	7.1	54.9	S 50	04 01	04 43	05 18	02 32	03 37	04 43	05 50
	02	206 36.2 26.3	222 28.7 12.0	19 12.1	7.1	54.8	52	03 52	04 37	05 14	02 22	03 29	04 37	05 45
	03	221 36.3 25.4	236 59.7 12.1	19 05.0	7.3	54.8	54	03 42	04 31	05 09	02 12	03 20	04 30	05 41
	04	236 36.3 24.5	251 30.8 12.2	18 57.7	7.3	54.8	56	03 31	04 23	05 04	02 00	03 10	04 23	05 35
	05	251 36.4 23.6	266 02.0 12.2	18 50.4	7.5	54.8	58	03 18	04 15	04 59	01 46	02 58	04 14	05 29
	06	266 36.5 S10 22.7	280 33.2 12.3	S18 42.9	7.5	54.8	S 60	03 02	04 05	04 52	01 29	02 45	04 04	05 22
	07	281 36.6 21.8	295 04.5 12.4	18 35.4	7.6	54.8	Lat.		SOL Pôr		CREP Civil Náut.		LUA - Pôr	
	08	296 36.6 20.8	309 35.9 12.4	18 27.8	7.7	54.8	h m		h m		h m		h m	
	09	311 36.7 19.9	324 07.3 12.4	18 20.1	7.7	54.7	h m		h m		h m		h m	
S A B A D O	10	326 36.8 19.0	338 38.7 12.6	18 12.4	7.9	54.7	N 72	16 12	17 25	18 44	13 23	15 27	17 15	17 00
	11	341 36.9 18.1	353 10.3 12.5	18 04.5	7.9	54.7	N 70	16 26	17 31	18 42	12 02	14 03	15 49	17 28
	12	356 36.9 S10 17.2	7 41.8 12.7	S17 56.6	8.0	54.7	68	16 38	17 36	18 41	12 51	14 31	16 06	17 38
	13	11 37.0 16.3	22 13.5 12.7	17 48.6	8.1	54.7	66	16 47	17 41	18 40	13 22	14 52	16 20	17 46
	14	26 37.1 15.4	36 45.2 12.8	17 40.5	8.1	54.7	64	16 55	17 44	18 40	13 46	15 08	16 32	17 53
	15	41 37.2 14.5	51 17.0 12.8	17 32.4	8.2	54.7	62	17 02	17 48	18 39	14 04	15 22	16 41	18 00
	16	56 37.3 13.6	65 48.8 12.9	17 24.2	8.3	54.6	60	17 08	17 51	18 39	14 19	15 34	16 50	18 05
	17	71 37.3 12.7	80 20.7 13.0	17 15.9	8.4	54.6	N 58	17 13	17 53	18 39	14 32	15 44	16 57	18 10
	18	86 37.4 S10 11.7	94 52.7 13.0	S17 07.5	8.4	54.6	56	17 18	17 56	18 39	14 44	15 53	17 04	18 14
	19	101 37.5 10.8	109 24.7 13.1	16 59.1	8.6	54.6	54	17 22	17 58	18 39	14 54	16 01	17 10	18 18
D O M I N G O	20	116 37.6 09.9	123 56.8 13.1	16 50.5	8.5	54.6	52	17 26	18 00	18 39	15 03	16 09	17 15	18 22
	21	131 37.6 09.0	138 28.9 13.2	16 42.0	8.7	54.6	50	17 29	18 02	18 40	15 13	16 24	17 27	18 29
	22	146 37.7 08.1	153 01.1 13.2	16 33.3	8.7	54.6	45	17 37	18 07	18 41	15 21	16 24	17 27	18 29
	23	161 37.8 07.2	167 33.3 13.3	16 24.6	8.8	54.6	N 40	17 43	18 11	18 42	15 37	16 37	17 36	18 35
	00	176 37.9 S10 06.3	182 05.6 13.4	S16 15.8	8.9	54.5	35	17 49	18 14	18 44	15 50	16 47	17 45	18 41
	01	191 38.0 05.4	196 38.0 13.4	16 06.9	8.9	54.5	30	17 54	18 18	18 46	16 01	16 57	17 52	18 46
	02	206 38.0 04.4	211 10.4 13.5	15 58.0	9.0	54.5	20	18 02	18 24	18 50	16 20	17 13	18 04	18 54
	03	221 38.1 03.5	225 42.9 13.6	15 49.0	9.1	54.5	N 10	18 10	18 31	18 56	16 36	17 27	18 15	19 01
	04	236 38.2 02.6	240 15.5 13.6	15 39.9	9.1	54.5	0	18 17	18 38	19 02	16 52	17 39	18 25	19 08
	05	251 38.3 01.7	254 48.1 13.6	15 30.8	9.2	54.5	S 10	18 24	18 46	19 11	17 07	17 52	18 34	19 14
D O M I N G O	06	266 38.4 S10 00.8	269 20.7 13.7	S15 21.6	9.2	54.5	20	18 32	18 55	19 21	17 24	18 06	18 45	19 21
	07	281 38.4 9 59.9	283 53.4 13.8	15 12.4	9.3	54.5	30	18 41	19 06	19 35	17 42	18 22	18 57	19 29
	08	296 38.5 59.0	298 26.2 13.8	15 03.1	9.4	54.4	35	18 47	19 13	19 44	17 53	18 31	19 04	19 34
	09	311 38.6 58.0	312 59.0 13.9	14 53.7	9.4	54.4	40	18 53	19 21	19 54	18 06	18 41	19 11	19 39
	10	326 38.7 57.1	327 31.9 13.9	14 44.3	9.5	54.4	45	19 00	19 30	20 08	18 20	18 53	19 20	19 45
	11	341 38.8 56.2	342 04.8 14.0	14 34.8	9.6	54.4	S 50	19 08	19 43	20 25	18 38	19 07	19 31	19 52
	12	356 38.9 S 9 55.3	356 37.8 14.1	S14 25.2	9.6	54.4	52	19 12	19 48	20 33	18 46	19 14	19 36	19 55
	13	11 38.9 54.4	11 10.9 14.0	14 15.6	9.6	54.4	54	19 17	19 55	20 43	18 56	19 21	19 42	19 59
	14	26 39.0 53.5	25 43.9 14.2	14 06.0	9.8	54.4	56	19 22	20 02	20 54	19 06	19 30	19 48	20 02
	15	41 39.1 52.5	40 17.1 14.2	13 56.2	9.7	54.4	58	19 27	20 10	21 06	19 18	19 39	19 55	20 07
D O M I N G O	16	56 39.2 51.6	54 50.3 14.3	13 46.5	9.9	54.4	S 60	19 33	20 20	21 22	19 32	19 50	20 02	20 12
	17	71 39.3 50.7	69 23.6 14.3	13 36.6	9.8	54.3	SOL		LUA		Pass Merid		Idade Fase	
	18	86 39.4 S 9 49.8	83 56.9 14.3	S13 26.8	10.0	54.3	ET (-)		Pass		Sup Inf		d %	
	19	101 39.4 48.9	98 30.2 14.4	13 16.8	10.0	54.3	00h 12h							